Customer Shopping Behavior Analysis

Uncovering insights from 3,900 transactions to guide strategic business decisions

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Project Overview

Dataset Scope

3,900 purchases across various product categories with 18 key features including customer demographics, purchase details, and shopping behavior.

Strategic Goals

Identify spending patterns, customer segments, product preferences, and subscription behavior to inform business strategy.

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Data Preparation Process

01

Data Loading

Imported dataset using pandas for analysis

02

Initial Exploration

Checked structure with df.info() and summary statistics with .describe()

03

Missing Data Handling

Imputed 37 missing values in Review Rating using median by category

04

Feature Engineering

Created age_group and purchase_frequency_days columns

05

Database Integration

Connected to PostgreSQL for SQL analysis

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Revenue by Gender

54%

Female Customers

Generated majority of revenue

46%

Male Customers

Contributed nearly half of total sales

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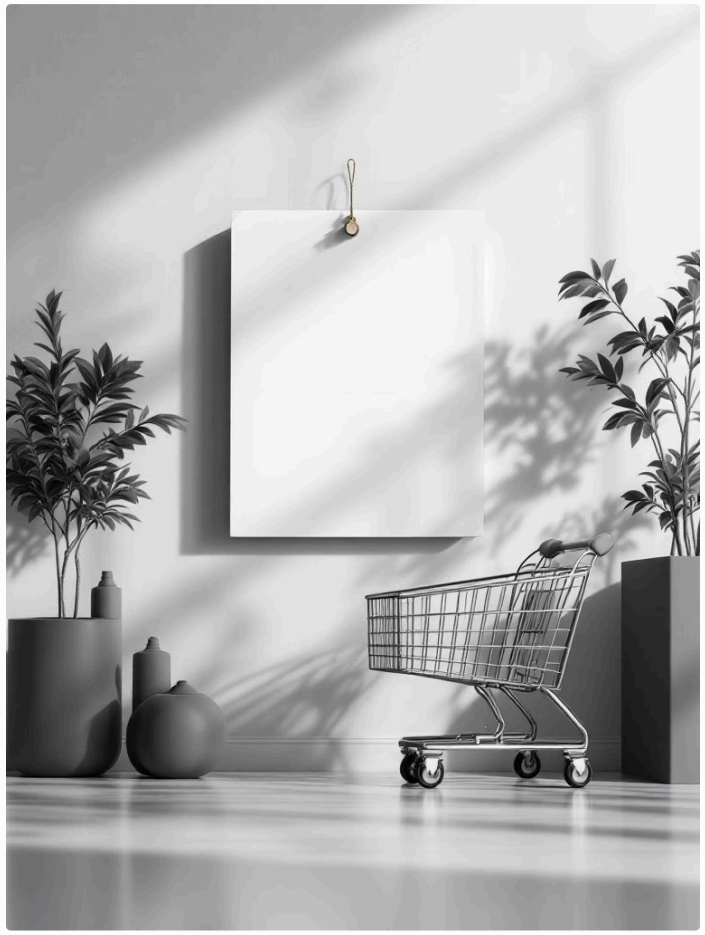
High-Spending Discount Users

Key Finding

Customers who used discounts but spent above average purchase amount represent valuable segment for targeted promotions.

Strategic Insight

These customers respond to incentives while maintaining high purchase values, indicating strong brand loyalty and willingness to spend.



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Top Products by Rating

Product A

4.8 average rating

Product B

4.7 average rating

Product C

4.6 average rating

Top-rated products should be highlighted in marketing campaigns to drive conversions.

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Shipping & Subscription Insights

Express Shipping

Higher average purchase amounts compared to Standard shipping, indicating premium customer segment

Subscribers

Higher average spend and total revenue vs. non-subscribers, demonstrating subscription value

Repeat Buyers

Customers with >5 purchases more likely to subscribe, showing loyalty progression

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Customer Segmentation

New

First-time buyers

Returning

Multiple purchases

Loyal

Consistent shoppers

Segmentation based on purchase history enables targeted retention strategies and personalized marketing.

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Revenue by Age Group

32%

35-50 Years

Highest revenue contribution

28%

25-34 Years

Second highest revenue

22%

51-65 Years

Steady spending pattern

18%

18-24 Years

Emerging customer base

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits and rewards for subscribers

Customer Loyalty Programs

Reward repeat buyers to move them into loyal segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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